Gradient Descent Approach to Inverse Kinematics

Summary:

This lesson introduces iterative (numerical) inverse kinematics. Instead of solving for the angles with triangles as in Lesson 6, the robot improves a guess: it uses forward kinematics to find where the tip lands, measures the error, and nudges every joint a little “downhill” so the error gets smaller. On Day 1, students do this by hand for one number, for one joint and for the flat arm. On Day 2, students extend the same loop to the 3D arm and study the results of a computer run: they choose a learning rate and see what happens with different start poses, targets that are out of reach and targets near full reach. At the end, students can explain how gradient descent IK works, check the final error, and compare it with the trigonometric method. All angles use the DH convention from Lessons 4 to 6.

Objective:

By the end of this lesson, students will be able to:

- Describe iterative IK as guess, measure the error, improve, and explain why it only needs forward kinematics
- Write the cost as the squared error in link lengths, and estimate a slope by nudging a joint
- Apply the step rule $\theta_new = \theta - \eta \times slope$ by hand to one number, one joint and the flat arm
- Extend gradient descent IK to the 3D arm and choose a sensible learning rate, tolerance and step limit
- Explain what happens when the target is out of reach, near full reach or approached from a different start pose, and always check the final error
- Compare gradient descent with the trigonometric method of Lesson 6

Standards:

NGSS

- Engineering Design: HS-ETS1-2 (break a complex problem into smaller problems), HS-ETS1-3 (evaluate solutions against trade-offs), HS-ETS1-4 (use a computer model to test a solution)
- Science & Engineering Practices: Developing and Using Models; Using Mathematics and Computational Thinking; Analyzing and Interpreting Data
- Crosscutting Concepts: Systems and System Models; Stability and Change

Vocabulary:

Iterative (numerical) IK: Solving IK by improving a guess again and again, instead of using a formula. It only needs forward kinematics.

Error: The distance from the tip of the arm to the target, in mm.

Cost: A score for a guess: $C = (\text{error} \div 250)^2$, the squared error measured in link lengths. The best angles make the cost as small as possible.

Slope: How much the cost changes when a joint turns a little. A positive slope means turning the joint up makes the cost bigger.

Gradient: The list of slopes, one for each joint (θB, θS and θE).

Gradient descent: Moving every angle a small step in the direction that lowers the cost, again and again: θ_new = θ − η × slope.

Learning rate (η, eta): The size of each step. Too small is slow; too big overshoots and bounces.

Nudging: Finding a slope without calculus: turn the joint a tiny amount h and see how the cost changes, (C(θ + h) − C(θ)) ÷ h.

Tolerance: How small the error must be before we stop. This lesson uses 0.5 mm.

Step limit: The most steps the loop may take (1000 here), so it can never run forever.

Start pose: The angles the loop begins from. Gradient descent rolls into the nearest valley, so the start can change which answer we get.

Radian: An angle unit where 180° = π ≈ 3.1416 rad. The step rule is used in radians.

Resources & Materials:

- Lesson 7 slideshow, Days 1 and 2 (36 slides; the slide numbers in the Procedures are from this slideshow)
- Calculators with sin and cos keys (degrees and radians modes) for the Day 1 Try It Yourself problems
- Students' answers from Lesson 6, for comparison
- The assembled 4-DOF arm from Lesson 2, to show reach and a fully straight arm (optional)
- All numbers use L1 = L2 = 250 mm and the DH angles from Lessons 4–6: θB is the base turn, θS is measured from the floor, θE is measured from the line of the upper arm, and up is +.

Procedures:

Day 1: Gradient Descent by Hand

- **Opening (5 min)** — Show the title slide, "Where Lesson 6 Left Off" and "Guess, Check, Improve" (slides 1–3). Triangles gave θB = 45°, θS = 45°, θE = −45° exactly, but only for this arm: every new joint needs new triangles. Today the robot guesses, uses FK to see where the tip lands, measures the error and nudges the angles, then repeats. Play "hot and cold" with a volunteer to motivate the loop.
- **Measuring the error (4 min)** — Show "Measuring the Error" (slide 4). The error is the distance from the tip to the target. The cost is C = (error ÷ 250)²: dividing by 250 mm measures the error in link lengths. Squared means it is never negative, it is zero only on the target, and it makes a smooth bowl. The goal is to make C as small as possible.
- **Rolling downhill (5 min)** — Show "Rolling Downhill" (slide 5). The cost plotted against an angle looks like a valley. Measure the slope where you stand, then step the opposite way: θ_new = θ −

η × slope. η (eta) is the learning rate, the size of each step. A negative slope makes θ bigger; a positive slope makes θ smaller.

- **Worked example: one number (5 min)** — Work through f(x) = (x − 3)² on slide 6, with slope 2(x − 3), η = 0.1 and x = 0 to start: 0.6, 1.08, 1.464, 1.7712, 2.017. Tell students the slope is given here; later they will measure it by nudging. Point out that the steps shrink as the slope flattens near the bottom.
- **Try It Yourself #1 (5 min)** — Students take two steps with η = 0.25 and with η = 1 (slide 7). Reveal (slide 8): 1.5 and 2.25 for (a), and 6, 0, 6, 0 … for (b). A step that is too big jumps over the bottom and never settles; keep this idea for Day 2.
- **One joint (5 min)** — Show "Gradient Descent on One Joint" (slide 9): one 250 mm link at angle θ from the floor and the target (250, 0), so the answer is θ = 0°. The tip is (250 cos θ, 250 sin θ), which gives C = 2 − 2 cos θ and slope = 2 sin θ. With η = 0.2 from 45°, fill in the table together: 28.79°, 17.76°, 10.77°, 6.49°, 3.90°. Teacher note: the step rule uses radians; change back to degrees at the end.
- **Measuring the slope by nudging (4 min)** — Show slide 10. Without calculus, nudge the angle by h = 0.01 and compute (C(θ + h) − C(θ)) ÷ h = (0.60000 − 0.58579) ÷ 0.01 = 1.42 at 45°, close to the exact 1.414. This only needs the cost, and the cost only needs FK, so it works for any arm.
- **Try It Yourself #2 (4 min)** — Students take two steps from θ = 30° with η = 0.2 (slide 11). Reveal 18.54° and 11.25° (slide 12).
- **The flat arm (3 min)** — Show "The Flat Arm: Two Joints" (slide 13). The cost now depends on θS and θE, and FK from Lesson 5 (flat_tip) gives the tip. The gradient is two slopes: nudge θS while keeping θE, then nudge θE while keeping θS. Both angles step downhill at the same time.
- **Worked example: (250, 250) (11 min)** — Slides 14–17. Start straight up at θS = 90°, θE = 0° (tip (0, 500)), with η = 0.1. The error is 353.6 mm, so C = 2. Nudging gives slopes ≈ 4 for θS and 2 for θE; both are positive, so both angles go down. One step gives θS = 1.1708 rad = 67.08° and θE = −0.2 rad = −11.46°, and the error falls from 353.6 mm to 187.0 mm. Show the table on slide 17: big steps at first, then smaller ones, and after 106 steps the error is under 0.5 mm at θS = 89.91°, θE = −89.85°, matching Lesson 6 (90°, −90°).
- **Same answer, no triangles (3 min)** — Show slide 18. Lesson 6 gives the exact answer in one pass but needs the triangles. Gradient descent gets very close and only needs FK and a loop. It trades exactness for generality.
- **Try It Yourself #3 (4 min)** — Students take one step for the target (341.51, 341.51) from θS = 90°, θE = 0°, using the given slopes 5.46 and 2.73 (slide 19). Reveal θS = 58.72°, θE = −15.64° (slide 20): the error falls from 376.5 mm to 51.8 mm in one step. Lesson 6's answer is (60°, −30°).
- **Recap (2 min)** — Show "Day 1 Recap" (slide 21): cost, slope, step, repeat. Preview Day 2: the base turns too, so there are three joints to nudge.

Day 2: The 3D Arm

- **Recap (3 min)** — Show "Day 2: Where Day 1 Left Off" (slide 22). Ask: What does η control? (The size of each step.) The slope is positive; does θ go up or down? (Down.) Why nudge instead of using a formula? (Nudging only needs FK, so it works for any arm.)
- **Extending to 3D (5 min)** — Show "Extending to 3D" (slide 23). Nothing in the method changes: tip_3d from Lesson 5 gives the tip, the cost is (error ÷ 250)², the gradient is three slopes (θB, θS and θE), and all three step downhill. By hand this is a lot of arithmetic, so a computer ran the

loop: for (301.78, 301.78, 176.78), starting at (0°, 45°, −45°) with η = 0.1, it stopped after 84 steps at (45.0°, 45.15°, −45.29°), 0.49 mm from the target and within 0.3° of Lesson 6's (45°, 45°, −45°). Students write this code themselves in Lesson 8.

- **Watching the error fall (6 min)** — Show slide 24. The error drops from 326.64 mm to 6.43 mm in 5 steps, then crawls: from step 20 it takes 64 more steps to get from 2.31 mm to 0.49 mm. The valley is flat near the bottom, so the slopes and the steps become tiny.
- **Choosing the learning rate (6 min)** — Show slide 25: η = 0.02 needs 364 steps, 0.05 needs 153, 0.1 needs 84, 0.2 needs 52, and 0.3 never settles (it bounces between 234.5 and 288.0 mm). Ask students which η they would choose and why. Use η = 0.1 as the safe default.
- **When do we stop? (3 min)** — Show slide 26: stop when the error is under the tolerance, or after the step limit. Stopping at the step limit does not mean the answer is right, so always check the final error.
- **Try It Yourself #4 (7 min)** — Students read a computer run for (295.75, 170.75, 91.51) (slide 27): it stopped after 79 steps at (30.0°, 59.91°, −89.85°) with an error of 0.49 mm. Did it reach the target? Reveal (slide 28): yes, the error is under the 0.5 mm tolerance, and the pose matches Lesson 6's (30°, 60°, −90°) to within 0.2°. With η = 0.2 the same run needs only 39 steps.
- **Different start, different answer (5 min)** — Show slide 29. For (−324.76, 187.50, 216.51) (Lesson 6's #5), start (0, 45, −45) finishes at [−30.0, 179.88, −59.78]: the base points away and the arm reaches back over the top. Start (150, 45, −45) finishes at [150.0, 59.88, −59.77], Lesson 6's answer. Both put the tip on the target. Gradient descent rolls into the nearest valley, so choose a start near the pose you want and check the joint limits.
- **Try It Yourself #5 (8 min)** — Students compare two computer runs for (−125, 216.51, 250) (slide 30): the start (0°, 45°, −45°) ends at (120.0°, 89.9°, −89.84°) in 73 steps, the same as Lesson 6, and the start (0°, 90°, 0°) ends at (−60.0°, 90.1°, 89.84°) in 106 steps. Are both right, and why did the base turn a different way? Reveal (slide 31): both errors are within the tolerance, so both tips are on the target. Starting straight up, the arm could fall either way, so it found the other valley. Choose the answer that fits the joint limits.
- **Out of reach (5 min)** — Show slide 32. For (400, 300, 200), trigonometry found no answer (the cosine came out as 1.32). Gradient descent does not: it uses all 1000 steps and stops with the arm straight (θE = 0°), pointing at the target, 38.52 mm short (538.52 − 500). Check the error to know the target was not reached.
- **When gradient descent struggles (3 min)** — Show the table on slide 33: out of reach (a leftover error), near full reach (very slow: (375, 216.51, 250) is still 1.5 mm off after 1000 steps, because the arm is almost straight and the slope is tiny), η too big (bounces), η too small (hundreds of steps), and a different start (a different valid answer).
- **Trigonometry vs. gradient descent (5 min)** — Show slide 34. Ask: if the arm had six joints, which method would you use, and why? With gradient descent the loop stays the same; only FK changes. Trigonometry is exact and fast but needs new triangles for every joint.
- **Making it faster (2 min)** — Show slide 35: warm starts from the current pose, exact slopes from calculus (Jacobian methods, Lesson 8) and adaptive steps. All are still guess, measure, improve. If the lesson runs long, mention this in one sentence.
- **Wrap-up (2 min)** — Show "What We Learned" (slide 36): iterative IK needs only FK; cost, slope and the step rule; the learning rate and the start pose matter; always check the final error. Preview Lesson 8: writing IK code and the Jacobian.

Assessment:

Students complete the Try It Yourself problems (#1–#5) and check each answer against the solution on the slide. On Day 2, pairs explain their answers to #4 and #5, using the final error of each run.

Use these questions as an exit ticket or short quiz. The answers are in red.

- For $f(x) = (x - 3)^2$ with slope $2(x - 3)$, start at $x = 0$ with $\eta = 0.25$. What is x after one step?
  Answer: $x = 0 - 0.25 \times (-6) = 1.5$.
- Why does the cost use the squared error instead of just the error?
  Answer: It is never negative, it is zero only when the tip is on the target, and it makes a smooth bowl to roll down.
- One link has cost $C = 2 - 2 \cos \theta$ and slope $2 \sin \theta$. Start at $\theta = 30°$ with $\eta = 0.2$. What is θ after one step?
  Answer: slope = $2 \sin 30° = 1$, so $\theta = 0.5236 - 0.2 \times 1 = 0.3236$ rad = 18.54°.
- What does the learning rate η control? What goes wrong if it is too big, and if it is too small?
  Answer: The size of each step. Too big: it overshoots and bounces without settling. Too small: it takes very many steps.
- How can a computer find the slope of the cost without calculus?
  Answer: Nudge the joint a tiny amount h and see how the cost changes: $(C(\theta + h) - C(\theta)) \div h$. This only needs FK.
- True or false: if gradient descent stops, the answer is correct.
  Answer: False. It may have hit the step limit, or the target may be out of reach. Always check the final error.
- The target (400, 300, 200) is out of reach. What does trigonometry do, and what does gradient descent do?
  Answer: Trigonometry finds $\cos \theta E = 1.32$ so there is no answer. Gradient descent ends at the closest point (arm straight, pointing at the target) with a leftover error of 38.52 mm.
- Two different start poses gave two different answers for the same target. Can both be right? Why?
  Answer: Yes, if both put the tip on the target. Gradient descent rolls into the nearest valley, so the start pose decides which valid answer we get. Check the joint limits before using one.
- In one or two sentences, why might an engineer choose gradient descent over trigonometry for a six-joint arm?
  Answer: It is the same loop for any arm: only the FK function changes, while trigonometry needs new triangles and special cases for every joint. The cost is some speed and exactness.